

Problem 3

W3 : Finding The Most Recent Purchase For Each SmartMart Customer

Problem Statement

SmartMart wants to analyze recent customer purchasing activity by identifying the **most recent purchase made by each customer**.

The query should retrieve the **customer name, purchase date, and purchase amount**, ensuring that only the latest purchase for each customer is included in the result.

This analysis will help SmartMart understand each customer's latest shopping activity and identify when customers last made a purchase.

Schema Details

Database: SmartMartDB

Customers Table

Column Name	Data Type	Description
customer_id	INT (PK)	Unique identifier for each customer
customer_name	VARCHAR(100)	Name of the customer
city	VARCHAR(50)	Customer's city

Purchases Table

Column Name	Data Type	Description
purchase_id	INT (PK)	Unique purchase identifier
customer_id	INT (FK)	Customer who made the purchase
purchase_date	DATE	Date when the purchase was made
purchase_amount	DECIMAL(10,2)	Total amount of the purchase

Sample Data

Customers Table

customer_id	customer_name	city
101	Rahul	Bangalore
102	Priya	Chennai
103	Arjun	Hyderabad
104	Sneha	Pune

Purchases Table

purchase_id	customer_id	purchase_date	purchase_amount
5001	101	2025-01-10	1250.00
5002	101	2025-02-15	1850.00
5003	101	2025-03-20	2200.00
5004	102	2025-01-05	950.00
5005	102	2025-02-25	1750.00
5006	103	2025-01-18	3200.00
5007	103	2025-03-12	2800.00
5008	103	2025-04-05	4500.00
5009	104	2025-02-10	1500.00
5010	104	2025-03-30	2100.00

Query to be Implemented

Retrieve the **most recent purchase for each SmartMart customer** using a **single SQL query**.

The solution must ensure:

- Each customer's latest purchase is correctly identified.
- Only the latest purchase of each customer is returned.
- The result should contain:
 - `customer_name`
 - `purchase_date`
 - `purchase_amount`

 **Tip**

Join both tables and use a correlated subquery with **MAX()** to find each customer's latest **purchase_date**.



Expected Output

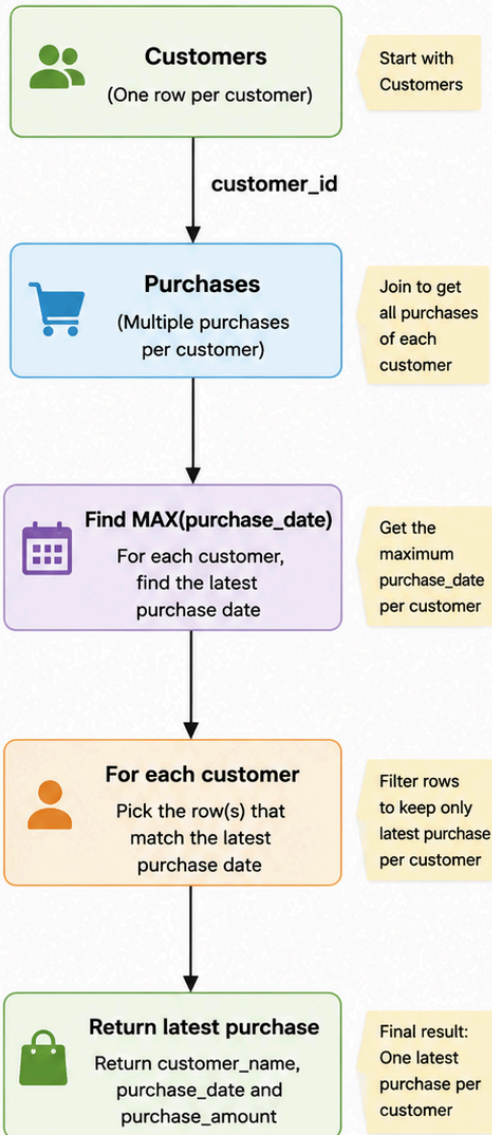
customer_name	purchase_date	purchase_amount
Rahul	2025-03-20	2200.00
Priya	2025-02-25	1750.00
Arjun	2025-04-05	4500.00
Sneha	2025-03-30	2100.00



SQL Concept Being Tested

LOGIC FLOW

Finding Most Recent Purchase
for Each Customer



EXPECTED SOLUTION PATTERN (Single Query)

```
SELECT
    c.customer_name,
    p.purchase_date,
    p.purchase_amount
FROM
    customers c
JOIN
    purchases p
ON c.customer_id = p.customer_id
WHERE
    p.purchase_date = (
        SELECT MAX(p2.purchase_date)
        FROM purchases p2
        WHERE p2.customer_id = c.customer_id
    );
```

How it works:

- 1 Join customers with purchases to get all purchases belonging to each customer.
- 2 The subquery (SELECT MAX(p2.purchase_date) ..) finds the latest **purchase_date** for the current customer.
- 3 The **WHERE** condition keeps only the rows where the **purchase_date** equals that latest date.
- 4 This ensures we get only the most recent purchase for each customer.
- 5 Finally, we return **customer_name**, **purchase_date** and **purchase_amount**.

💡 Tip

If multiple purchases exist on the same latest date for a customer, this query will return all of them.